

|          |                                                                               |
|----------|-------------------------------------------------------------------------------|
| Document | P1732R0                                                                       |
| Date     | 2019-06-14                                                                    |
| Author   | CJ Johnson < <a href="mailto:johnsoncj@google.com">johnsoncj@google.com</a> > |
| Audience | Library Evolution Working Group (LEWG), Library Working Group (LWG)           |

## Do not promise support for function syntax of operators

From Hyrum's Law [\[1\]](#):

With a sufficient number of users of an API,  
it does not matter what you promise in the contract:  
all observable behaviors of your system  
will be depended on by somebody.

### Proposal

In the Library Evolution Working Group (LEWG) room of the Kona 2019 ISO C++ meeting, it was mentioned that the only supported way to invoke operators on types in the standard library is operator syntax. Function syntax is not supported. This paper proposes updating SD-8 to convey this information to users of the standard library.

For clarity...

Unsupported: `x.operator+(y)`, `operator+(x, y)`

Supported: `x + y`

### Wording option 1

Primarily, the standard reserves the right to:

[...]

- \* Make changes to existing interfaces in a fashion that will be backward compatible, if those interfaces are solely used to instantiate types and invoke functions. Implementation details (the primary name of a type, the implementation details for a function callable
- + , the way an operator is or is not overloaded
- ) may not be depended upon.

[...]

### Wording option 2

Primarily, the standard reserves the right to:

[...]

- + \* Change the way operators are defined on types in the standard library
- + (between not overloaded, member overloaded and non-member overloaded).

[...]

## Wording option 3

Primarily, the standard reserves the right to:

[...]

+ \* Assume operators are only invoked using operator syntax (not function  
+ syntax).

[...]

## References

[1] [Hyrum's Law](#)